

Report by:

- Dr. Tripura Pidikiti, Hyderabad Hub Leader, IEEE IES Hubs and Nodes Initiative
- Dr. Deepak Ronaki, Organiser, Assistant Professor, IITM, Madras
- Mr. Anakapalli Vasu Deva, Research Scholar, IITM, Madras

Workshop on EV Battery and Charging Technology

Dates: January 5–6, 2026

Venue: Room No. 203, NAC–2, Indian Institute of Technology Madras (IIT Madras), India

Organised by: Department of Engineering Design, IIT Madras

In association with: IEEE Industrial Electronics Society (IES) – HUBS & NODES Initiative

1. Overview and Objectives

The two-day **Workshop on EV Battery and Charging Technology** was organised at the Indian Institute of Technology Madras on January 5–6, 2026, with the objective of providing an in-depth technical and system-level understanding of **electric vehicle (EV) batteries, charging infrastructure, power electronics, wireless charging, and vehicle-to-grid (V2G) technologies**.

The workshop served as a **platform for interaction between academia, industry professionals, researchers, and students**, while also marking a significant milestone as the **first IEEE IES HUBS & NODES activity conducted in the Chennai region**. The program aligned with IEEE IES's mission of strengthening **industry–academia collaboration, professional development, and dissemination of cutting-edge industrial electronics knowledge**.

This event represents the **first officially conducted IEEE IES HUBS & NODES initiative activity in the Chennai region**.

1.1 IEEE IES Support and Sponsorship

The workshop was supported under the **IEEE Industrial Electronics Society (IES) HUBS & NODES Initiative**, with partial financial assistance provided by IEEE IES. This support enabled broader participation, high-quality technical sessions, and strengthened engagement between academia, industry, and the IEEE IES community.

2. Participation Overview

2.1 Participant Summary

- **Total Participants:** 89
- **Industry Professionals:** 24
- **Faculty Members:** 4
- **Students & Young Professionals:** 61

Participants represented IIT Madras and several other academic, research, and industrial organisations, enabling interdisciplinary interaction and knowledge exchange.

2.2 Participant Demographics

Category	Number
Undergraduate Students (UG)	11
Postgraduate Students (PG)	20
Doctoral Research Scholars (PhD)	30
Faculty Members	4
Industry Professionals	24
Total	89

The balanced participation across **UG, PG, PhD scholars, faculty, and industry professionals** facilitated rich technical discussions, interdisciplinary knowledge exchange, and meaningful interaction between research, education, and practical deployment perspectives in electric vehicle and charging technologies.

3. Inauguration Ceremony (Day 1: 8:40 AM – 9:30 AM)

3.1 Dignitaries

- **Chief Dignitary:** Dr. C. S. Shankar Ram
- **Chief Guest:** Dr. Nilesh J. Vasa

- **Dais Member:** Dr. Tripura Pidikiti, IEEE IES Standing Committee Member (MDC), IEEE IES AdCom Member-at-Large.

3.2 Resource Persons

- Dr. Gautham Ram Chandra Mouli, Associate Professor of Electric Mobility, TU Delft, Netherlands
- Dr. Vishnu Mahadeva Iyer, Assistant Professor, IISc Bangalore
- Dr. Deepak Ronanki, Assistant Professor, Department of Engineering Design, IIT Madras (Workshop Organiser)

Lamp Lighting and Invocation

The workshop commenced with a traditional lamp-lighting ceremony, symbolising the pursuit of knowledge and innovation, followed by an invocation prayer.

Welcome Address

Dr. Deepak Ronanki welcomed the dignitaries, speakers, and participants, emphasising the significance of the event as the **first IEEE IES HUBS & NODES initiative in Chennai**, and highlighting the importance of collaborative engagement in the rapidly evolving EV ecosystem.

3.3 Key Messages and Insights

- Dr. C. S. Shankar Ram felicitated the resource persons and highlighted the Department of Engineering Design's initiatives in **power electronics, control systems, and e-mobility research**.
- Dr. Nilesh J. Vasa spoke on the **global energy transition**, stressing:
 - The limited long-term economic viability of oil dependency (25–30 years)
 - The urgent environmental need for sustainable mobility
 - The strategic importance of e-mobility for economic growth
- Dr. Gautham Ram Chandra Mouli highlighted the importance of a **well-to-wheel approach** in EV planning, stressing that the **source of electricity generation** is critical to the true sustainability of EVs.
- Dr. Vishnu Mahadeva Iyer presented comparative statistics on global charging infrastructure, pointing out India's urgent need for expansion, standardisation, and interoperability.

- Dr. Tripura Pidikiti presented an overview of **IEEE IES initiatives**, including the **HUBS & NODES model**, and discussed their role in strengthening local engagement with global networks.

4. Technical Program – Day 1

4.1 Session 1: EV Battery Technology

Speaker: Dr. Gautham Ram Chandra Mouli

This session provided a comprehensive overview of **battery chemistry, pack design, thermal management, and ageing mechanisms**. Key discussions included:

- Comparison of energy densities across Li-ion, Li-S, Li-air, sodium-ion, and silicon-based batteries
- Trade-offs in **cell form factors** (18650, 2170, 4680, prismatic, pouch)
- Battery Management Systems (BMS) and Thermal Management Systems (TMS)
- Internal resistance effects on EV performance, fast charging, and ageing
- Cost structure, recycling, and circularity of Li-ion batteries
- Design considerations illustrated through real-world EV examples

The session emphasised the **interdependence of chemistry, thermal control, safety, cost, and lifecycle performance** in modern EV battery systems.

4.2 Session 2: Multiport Chargers for Electric 2W/3W Vehicles

Speaker: Dr. Vishnu Mahadeva Iyer

This session addressed **power electronics solutions** for India's fragmented EV ecosystem. Topics included:

- Multiport AC–DC PFC converter topologies
- Power factor correction and harmonic mitigation
- Independent CC–CV control for multiple battery packs
- Experimental validation demonstrating high efficiency, low THD, and robust performance

The approach demonstrated how **multiport chargers can address non-standardised battery systems** and enable scalable infrastructure.

4.3 Session 3: Reconfigurable Power Electronics for Scalable EV Charging Hubs

The discussion focused on **three-phase PFC architectures**, soft-switching techniques, and reconfigurable outputs enabling:

- Multiple low-voltage charging ports for light EVs
- High-voltage charging for passenger cars
- Efficient and flexible charging hub designs

4.4 Session 4: Charging Technology for Heavy-Duty Vehicles

Speaker: Dr. Gautham Ram Chandra Mouli

This session examined **megawatt-scale charging solutions** for trucks, buses, ships, and aircraft. Key highlights:

- Emissions contribution of heavy-duty vehicles
- EU truck classifications (N1, N2, N3) and corresponding charging requirements
- Megawatt Charging System (MCS), pantograph charging, and battery swapping
- Case studies on electric trucks, ships, ferries, mining vehicles, and aircraft
- Charging corridors and global deployment initiatives

5. IEEE IES Virtual Engagement – Day 1

Speaker: Dr. Juan José Rodríguez-Andina

Role: IEEE IES President (2026–2027)

Participants interacted with **Dr. Juan José Rodríguez-Andina**, IEEE IES President (2026–2027), who shared insights into:

- IEEE IES global initiatives
- Conferences and professional development opportunities
- The strategic vision of the HUBS & NODES initiative

6. Laboratory Visits (Day 1)

Participants visited multiple laboratories at IIT Madras, including:

- Battery Charging Technology Lab
- Battery Engineering Technology Lab
- Automotive Systems Lab
- Motors and Control Lab

These visits bridged **theoretical concepts with experimental and system-level implementations**.

7. Technical Program – Day 2

7.1 Session 6: Wireless Charging Technology for EVs

Speaker: Dr. Deepak Ronanki

This session covered:

- Static and dynamic wireless charging
- Coil design, power electronics, and control
- Challenges in efficiency, alignment, EMC, and cost
- Modular and opportunity charging concepts
- Single-stage vs dual-stage architectures

7.2 Session 7: Vehicle-to-Grid (V2G) Technology

Speaker: Dr. Gautham Ram Chandra Mouli

The session provided a **quantitative and system-level analysis of V2G**, covering:

- Bidirectional charging architectures
- Grid services enabled by EV fleets
- Battery degradation considerations
- Case studies from Texas (USA) and the Netherlands
- Standards such as CHAdeMO and ISO 15118

8. IEEE IES Virtual Engagement – Day 2

Speaker: Dr. Milos Manic

Role: IEEE IES President (2024-2025), addressed participants on:

- IEEE IES membership activities
- Global technical committees
- The vision of being “globally connected and locally empowered”

9. Closing Ceremony

The closing ceremony featured:

- **Chief Guest:** Dr. Atriya Biswas, Assistant Professor, Department of Engineering Design, IIT Madras
- Reflections from resource persons and organisers
- Participant feedback and certificate distribution

Dr. Tripura Pidikiti highlighted the **strategic role of IEEE IES** in shaping future industrial electronics research and reinforced the importance of **academia–industry collaboration** in EV technologies.

10. Research Park Visit – IIT Madras

Participants visited the **IIT Madras Research Park**, interacting with startups, battery research centres, and Centres of Excellence focused on EVs and energy storage, gaining exposure to **technology commercialisation and translational research**.

11. Industry Participation

11.1 Industry Participants – Organisation-wise

The workshop saw strong participation from industry professionals representing leading organisations across **EVs, power electronics, automation, control systems, and digital engineering**. The industry participants and their designations are listed below:

Sl. No.	Designation	Organisation
1	Staff Engineer	Applied Materials India Pvt. Ltd.
2	Control Systems Engineer	Wabtec
3	Electrical Engineer	Trane Technologies
4	Senior Consultant	Tata Consultancy Services (TCS)
5	Senior Consultant	Tata Consultancy Services (TCS)
6	Assistant Consultant	Tata Consultancy Services Ltd.
7	Product Owner	Tata Consultancy Services, Chennai
8	Specialist	Tata Elxsi
9	Senior Engineer	Tata Elxsi
10	Senior Engineer	Tata Elxsi Limited

11	Deputy Manager	Stellantis
12	Director	Melcon Controls Private Limited
13	Product Engineer	Velankani Information Systems Pvt. Ltd.
14	Assistant Manager	Amara Raja Advanced Cell Technologies Limited
15	Engineer	Magna
16	Motors and Drives Engineer	Atlas Copco

11.2 Industry Representation Summary

The presence of professionals from **Applied Materials, Tata Group companies (TCS, Tata Elxsi), Stellantis, Magna, Atlas Copco, Amara Raja Advanced Cell Technologies, Trane Technologies, and Wabtec** significantly enriched discussions by bringing **practical deployment perspectives, industrial challenges, and real-world constraints** into advanced topics such as EV batteries, charging infrastructure, and power electronics.

12. Feedback and Outcomes

- High appreciation for technical depth
- Strong interest in partial power processing
- Demand for longer, hands-on workshops
- Enthusiasm for future IEEE IES initiatives

12.1 IEEE IES Outreach Impact

The workshop significantly enhanced awareness of IEEE IES activities among participants. Several students and young professionals expressed interest in IEEE IES membership, volunteering, and future chapter-level activities, indicating strong potential for sustained engagement in the Chennai region.

13. Conclusion

The workshop successfully delivered **state-of-the-art knowledge** on EV batteries and charging technologies, fostered **active interaction between academia and industry**, and reinforced IEEE IES's role in advancing industrial electronics education and innovation. The workshop also marked the beginning of sustained IEEE IES engagement in Chennai through the HUBS & NODES framework, with future technical programs, student involvement, and chapter-level activities planned. The event set a strong foundation for **future IEEE IES activities in the region** under the HUBS & NODES initiative.

